

Systems of Linear Equations

Chapter 7

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Outline of Lecture

Matrices and Echelon Form

Elementary Row Operations

Gaussian Elimination

Gauss-Jordan Elimination

Homogeneous Systems

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Matrices and Echelon Form

Matrices and Echelon Form (1 of 1)

Definition

A matrix is in **row echelon form** if it satisfies the following properties:

1. Any rows consisting entirely of zeros are at the bottom.
2. In each nonzero row, the first nonzero entry (called the **leading entry**) is in a column to the left of any leading entries below it.

Example 2.8

Assuming that each of the following matrices is an augmented matrix, write out the corresponding systems of linear equations and solve them.

$$\begin{bmatrix} 2 & 4 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 2 & 0 & 1 & -1 & 3 \\ 0 & 0 & -1 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 \end{bmatrix}$$

Solution:

We first remind ourselves that the last column in an augmented matrix is the vector of constant terms. The first matrix then corresponds to the system

$$2x_1 + 4x_2 = 1$$

$$-x_2 = 2$$

Example 2.8 – Solution (1 of 2)

Notice that we have dropped the last equation $0 = 0$, or $0x_1 + 0x_2 = 0$, which is clearly satisfied for any values of x_1 and x_2 .) Back substitution gives $x_2 = -2$ and then

$2x_1 = 1 - 4(-2) = 9$, so $x_1 = \frac{9}{2}$. The solution is $[\frac{9}{2}, -2]$.

The second matrix has the corresponding system

$$x_1 = 1$$

$$x_2 = 5$$

$$0 = 4$$

Example 2.8 – Solution (2 of 2)

The last equation represents $0x_1 + 0x_2 = 4$, which clearly has no solutions. Therefore, the system has no solutions. Similarly, the system corresponding to the fourth matrix has no solutions. For the system corresponding to the third matrix, we have

$$x_1 + x_2 + 2x_3 = 1$$

$$x_3 = 3$$

so $x_1 = 1 - 2(3) - x_2 = -5 - x_2$. There are infinitely many solutions, since we may assign x_2 any value t to get the parametric solution $[-5 - t, t, 3]$.

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Elementary Row Operations

Elementary Row Operations (1 of 2)

Definition

The following ***elementary row operations*** can be performed on a matrix:

1. Interchange two rows.
2. Multiply a row by a nonzero constant.
3. Add a multiple of a row to another row.

The process of applying elementary row operations to bring a matrix into row echelon form, called ***row reduction***, is used to reduce a matrix to echelon form.

Example 2.9

Reduce the following matrix to echelon form:

$$\begin{bmatrix} 1 & 2 & -4 & -4 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 3 & 2 & 1 & 5 \\ -1 & 1 & 3 & 6 & 5 \end{bmatrix}$$

Example 2.9 – Solution (1 of 5)

We work column by column, from left to right and from top to bottom. The strategy is to create a leading entry in a column and then use it to create zeros below it.

The entry chosen to become a leading entry is called a ***pivot***, and this phase of the process is called ***pivoting***. Although not strictly necessary, it is often convenient to use the second elementary row operation to make each leading entry a 1.

Example 2.9 – Solution (2 of 5)

We begin by introducing zeros into the first column below the leading 1 in the first row:

$$\begin{bmatrix} 1 & 2 & -4 & -4 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 3 & 2 & 1 & 5 \\ -1 & 1 & 3 & 6 & 5 \end{bmatrix} \xrightarrow[\begin{smallmatrix} R_2 - 2R_1 \\ R_3 - 2R_1 \\ R_4 + R_1 \end{smallmatrix}]{\text{blue arrow}} \begin{bmatrix} 1 & 2 & -4 & -4 & 5 \\ 0 & 0 & 8 & 8 & -8 \\ 0 & -1 & 10 & 9 & -5 \\ 0 & 3 & -1 & 2 & 10 \end{bmatrix}$$

The first column is now as we want it, so the next thing to do is to create a leading entry in the second row, aiming for the staircase pattern of echelon form.

Example 2.9 – Solution (3 of 5)

In this case, we do this by interchanging rows. (We could also add row 3 or row 4 to row 2.)

$$\begin{array}{c} R_2 \leftrightarrow R_3 \\ \longrightarrow \end{array} \begin{bmatrix} 1 & 2 & -4 & -4 & 5 \\ 0 & -1 & 10 & 9 & -5 \\ 0 & 0 & 8 & 8 & -8 \\ 0 & 3 & -1 & 2 & 10 \end{bmatrix}$$

The pivot this time was -1 .

Example 2.9 – Solution (4 of 5)

We now create a zero at the bottom of column 2, using the leading entry -1 in row 2:

$$\xrightarrow{R_4 + 3R_2} \begin{bmatrix} 1 & 2 & -4 & -4 & 5 \\ 0 & -1 & 10 & 9 & -5 \\ 0 & 0 & 8 & 8 & -8 \\ 0 & 0 & 29 & 29 & -5 \end{bmatrix}$$

Column 2 is now done. Noting that we already have a leading entry in column 3, we just pivot on the 8 to introduce a zero below it. This is easiest if we first divide row 3 by 8:

$$\xrightarrow{\frac{1}{8}R_3} \begin{bmatrix} 1 & 2 & -4 & -4 & 5 \\ 0 & -1 & 10 & 9 & -5 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 29 & 29 & -5 \end{bmatrix}$$

Example 2.9 – Solution (5 of 5)

Now we use the leading 1 in row 3 to create a zero below it:

$$\xrightarrow{R_4 - 29R_3} \begin{bmatrix} 1 & 2 & -4 & -4 & 5 \\ 0 & -1 & 10 & 9 & -5 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 24 \end{bmatrix}$$

With this final step, we have reduced our matrix to echelon form.

Elementary Row Operations (2 of 2)

Definition

Matrices A and B are **row equivalent** if there is a sequence of elementary row operations that converts A into B .

Theorem 2.1

Matrices A and B are row equivalent if and only if they can be reduced to the same row echelon form.

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Gaussian Elimination

Gaussian Elimination (1 of 2)

Gaussian Elimination

1. Write the augmented matrix of the system of linear equations.
2. Use elementary row operations to reduce the augmented matrix to row echelon form.
3. Using back substitution, solve the equivalent system that corresponds to the row-reduced matrix.

Example 2.10

Solve the system

$$2x_2 + 3x_3 = 8$$

$$2x_1 + 3x_2 + x_3 = 5$$

$$x_1 - x_2 - 2x_3 = -5$$

Solution:

The augmented matrix is

$$\left[\begin{array}{ccc|c} 0 & 2 & 3 & 8 \\ 2 & 3 & 1 & 5 \\ 1 & -1 & -2 & -5 \end{array} \right]$$

Example 2.10 – Solution (1 of 4)

We proceed to reduce this matrix to row echelon form, following the guidelines given for step 2 of the process. The first nonzero column is column 1. We begin by creating a leading entry at the top of this column; interchanging rows 1 and 3 is the best way to achieve this.

$$\left[\begin{array}{ccc|c} 0 & 2 & 3 & 8 \\ 2 & 3 & 1 & 5 \\ 1 & -1 & -2 & -5 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -5 \\ 2 & 3 & 1 & 5 \\ 0 & 2 & 3 & 8 \end{array} \right]$$

Example 2.10 – Solution (2 of 4)

We now create a second zero in the first column, using the leading 1:

$$\xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -5 \\ 0 & 5 & 5 & 15 \\ 0 & 2 & 3 & 8 \end{array} \right]$$

We now cover up the first row and repeat the procedure. The second column is the first nonzero column of the submatrix. Multiplying row 2 by $\frac{1}{5}$ will create a leading 1.

$$\left[\begin{array}{ccc|c} 1 & -1 & -2 & -5 \\ 0 & 5 & 5 & 15 \\ 0 & 2 & 3 & 8 \end{array} \right] \xrightarrow{\frac{1}{5}R_2} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -5 \\ 0 & 1 & 1 & 3 \\ 0 & 2 & 3 & 8 \end{array} \right]$$

Example 2.10 – Solution (3 of 4)

We now need another zero at the bottom of column 2:

$$\xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & -1 & -2 & -5 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

The augmented matrix is now in row echelon form, and we move to step 3.

Example 2.10 – Solution (4 of 4)

The corresponding system is

$$x_1 - x_2 - 2x_3 = 25$$

$$x_2 + x_3 = 3$$

$$x_3 = 2$$

and back substitution gives $x_3 = 2$, then $x_2 = 3 - x_3 = 3 - 2 = 1$, and finally $x_1 = -5 + x_2 + 2x_3 = -5 + 1 + 4 = 0$. We write the solution in vector form as

$$\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

Gaussian Elimination (2 of 2)

Definition

The **rank** of a matrix is the number of nonzero rows in its row echelon form.

Theorem 2.2

The Rank Theorem

Let A be the coefficient matrix of a system of linear equations with n variables. If the system is consistent, then

$$\text{number of free variables} = n - \text{rank}(A)$$

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Gauss-Jordan Elimination

Gauss-Jordan Elimination (1 of 2)

Definition

A matrix is in **reduced row echelon form** if it satisfies the following properties:

1. It is in row echelon form.
2. The leading entry in each nonzero row is a 1 (called a **leading 1**).
3. Each column containing a leading 1 has zeros everywhere else.

Gauss-Jordan Elimination (2 of 2)

Gauss-Jordan Elimination

1. Write the augmented matrix of the system of linear equations.
2. Use elementary row operations to reduce the augmented matrix to reduced row echelon form.
3. If the resulting system is consistent, solve for the leading variables in terms of any remaining free variables.

Example 2.13

Solve the system

$$w - x - y + 2z = 1$$

$$2w - 2x - y + 3z = 3$$

$$-w + x - y = -3$$

by Gauss-Jordan elimination.

Solution:

The reduction proceeds until we reach the echelon form:

$$\left[\begin{array}{cccc|c} 1 & -1 & -1 & 2 & 1 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Example 2.13 – Solution (1 of 2)

We now must create a zero above the leading 1 in the second row, third column. We do this by adding row 2 to row 1 to obtain

$$\left[\begin{array}{cccc|c} 1 & -1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

The system has now been reduced to

$$w - x + z = 2$$

$$y - z = 1$$

Example 2.13 – Solution (2 of 2)

It is now much easier to solve for the leading variables:

$$w = 2 + x - z \quad \text{and} \quad y = 1 + z$$

If we assign parameters $x = s$ and $z = t$ as before, the solution can be written in vector form as

$$\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 + s - t \\ s \\ 1 + t \\ t \end{bmatrix}$$

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Homogeneous Systems

Homogeneous Systems (1 of 1)

Definition

A system of linear equations is called **homogeneous** if the constant term in each equation is zero.

Theorem 2.3

If $[A \mid \mathbf{0}]$ is a homogeneous system of m linear equations with n variables, where $m < n$, then the system has infinitely many solutions.

The End of Chapter 7